The SSL Challenge: Report

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Problem Statement: Task provided is to use self supervised learning techniques and find which approach provides better accuracy on a dataset provided of approx 2400 samples spread across 16 classes.

Repo link: https://github.com/osavlani/ITCS6169-A2

Self supervised learning is a type of training where instead of hard labels, the model tries to learn semantic features of the data from the data itself. A task can try to target a particular property of the image, where the property can be a general feature of the image with the answer being contained in the data itself. Such methods don't often require labels and have many different architectures such as SimCLR [1], MoCo [2], BYOL [3], SimSiam [4], DINO [5], MAE [6], BEiT [7], Joint/multi-task, Autoencoders, VAE's [8] etc. For my testing I choose to implement MoCo, MAE, DINO and a multi-task approach with rotation [9] being my auxiliary task.

Alternate approaches like SimCLR and many more were also applicable but were not feasible due to certain reasons. SimCLR employs a huge batch size and requires large memory to process said batch size which wasn't available locally. VAE and SimSiam were feasible implementations but different approaches from their category were selected, MAE and DINO. Transformer architecture [10] could also be employed with these approaches but due to the dataset being small CNN seemed a better option. For reference we had two randomly initialized ResNet18 [11] CNN's, CNN with no training gave **41%** kNN accuracy and CNN with supervised training techniques gave **77.8%** validation accuracy. CNN initialized with training would be considered our baseline, and with strong regularization consisting of SGD optimizer, cosine LR [12], label smoothing [13], Mixup/CutMix [14, 15] our baseline is very competitive.

MoCo (Momentum Contrast) [2] was the first approach. A query encoder and a key encoder get different augmentations of the same sample, and the query has to pick its true match out of a queue of negative keys. The true match sits at position 0, and after a softmax the probability on position 0 gives the contrastive loss. Only the query encoder is trained by backpropagation, and the key encoder follows it as a momentum average. The pretrained weights are then reused to fine tune on labelled data. MoCo gave 83.2% validation accuracy, better than the baseline given the limited data, and more optimal hyperparameters might add another 2-5%.

The next approach tested was MAE, also known as masked autoencoders [6]. In this approach 75% of the image is masked (mean-filled) before entering the encoder, from the remaining features the decoder has to learn to fill the image(holes) and the loss is computed on the fill of those masked patches. MSE is taken of the real pixels against the fill of the masked patches. The encoder and decoder train together through back propagation and then the pretraining finishes. The encoder is carried forward to fine tuning where it is trained on labelled classes. The accuracy

of the implementation turned out to be **79.16%** validation accuracy which wasn't a significant gain but maybe with a different masking ratio it could provide a significant gain.

The third experiment was DINO [5], a student-teacher setup based on self distillation with no labels. Each image is cropped several times, two large global crops and two small local crops. The teacher sees only the global crops, while the student sees all of them. Both outputs become a probability distribution over 2,048 learned prototypes, and the loss is the cross entropy between the teacher's and the student's. The student therefore has to predict from a small crop what the teacher sees in the whole image. Only the student is trained by backpropagation, and the teacher is its exponential moving average. The teacher backbone is carried forward to fine tuning, with a kNN check every 5 epochs to pick the best one. DINO reached **85.81%** validation accuracy, 8 points above the baseline run, and tuning or longer pretraining could widen the gap.

The final experiment was Joint SSL + supervised. MoCo, DINO and MAE first train on a pretext task and then fine tune with labels, whereas the joint approach trains both tasks at the same time. Classification is the main task and rotation prediction [9] is the auxiliary one. The classification head classifies each sample, while a rotation head predicts the rotation (0, 90, 180 or 270 degrees) applied to copies of it. The two losses are summed, with the rotation loss weighted 0.5, for backpropagation, and only the classifier head is kept afterward. This gave **84.5%** validation accuracy. Other auxiliary tasks such as inpainting or colorization might suit the scene based dataset better.

Final conclusion, DINO performed best with 85.81% validation accuracy and the highest test accuracy of **77.3%**. Most of the approaches had an early stopping during training meaning with a better hyper parameter set maybe more features could have been learnt. Self supervised learning is best suited for large sets of unlabelled data [16] and considering dataset being limited to ~2k samples gains were seen but they could have been even significant provided more data. I personally preferred joint SSL over DINO as the auxiliary loss also shapes the shared backbone making the network learn semantic features through a weighted coefficient which we can adjust to decide how much the classification should be based on the semantic learnings.

Method	Val Accuracy	Test Accuracy
Supervised Baseline	77.8%	69.0%
SSL + Fine-tuning (MoCo)	83.2%	75.5%
SSL + Fine-tuning (MAE-style)	79.2%	73.0%
SSL + Fine-tuning (DINO)	85.8%	77.3%
Joint SSL + Classification	84.5%	74.3%
Best Model (DINO + fine-tune)	85.8%	77.3%

References

[1] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in ICML, 2020. arXiv:2002.05709.

[2] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in CVPR, 2020. arXiv:1911.05722.

[3] J.-B. Grill et al., "Bootstrap your own latent: A new approach to self-supervised learning," in NeurIPS, 2020. arXiv:2006.07733.

[4] X. Chen and K. He, "Exploring simple Siamese representation learning," in CVPR, 2021. arXiv:2011.10566.

[5] M. Caron et al., "Emerging properties in self-supervised vision transformers," in ICCV, 2021. arXiv:2104.14294.

[6] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in CVPR, 2022. arXiv:2111.06377.

[7] H. Bao, L. Dong, S. Piao, and F. Wei, "BEiT: BERT pre-training of image transformers," in ICLR, 2022. arXiv:2106.08254.

[8] D. P. Kingma and M. Welling, "Auto-encoding variational Bayes," in ICLR, 2014. arXiv:1312.6114.

[9] S. Gidaris, P. Singh, and N. Komodakis, "Unsupervised representation learning by predicting image rotations," in ICLR, 2018. arXiv:1803.07728.

[10] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in ICLR, 2021. arXiv:2010.11929.

[11] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in CVPR, 2016. arXiv:1512.03385.

[12] I. Loshchilov and F. Hutter, "SGDR: Stochastic gradient descent with warm restarts," in ICLR, 2017. arXiv:1608.03983.

[13] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the Inception architecture for computer vision," in CVPR, 2016. arXiv:1512.00567.

[14] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in ICLR, 2018. arXiv:1710.09412.

[15] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," in ICCV, 2019. arXiv:1905.04899.

[16] E. Cole, X. Yang, K. Wilber, O. Mac Aodha, and S. Belongie, "When does contrastive visual representation learning work?" arXiv:2105.05837, 2021.